



Build Pixel's Script

Quick facilitation notes & answer key

Goal

Students WRITE code for the first time in this subject (C3.1): Bit's script is given, and they author Pixel's script so the two sprites MEET when one green flag starts both. They must first read Bit's result, then create a script that produces a matching concurrent outcome. Writing a second script to coordinate with a running first script is the create-concurrent-code skill in C3.1.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a strip of move-blocks (forward 1/2/3, back 1/2/3) students can place in Pixel's blank slot before writing it down.

Run it (about 20 min)

1. I DO: read Bit's given script — 'start 0, forward 3 → 3.' Bit lands on 3. Write that target on the board.
2. WE DO: Pixel starts at 5 — ask 'what one move gets Pixel from 5 to 3?' Count back: back 2.
3. YOU DO: students write Pixel's move (back 2), then press the green flag and check both land on 3.
4. Connect: 'You wrote a script so two sprites running at once meet — that's writing concurrent code.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 3 → 3. Bit lands on 3.

Ex 2 — Pixel starts at 5 and must reach 3, so the move is back 2. (Any script that lands Pixel on 3 is correct; back 2 is the simplest.)

Ex 3 — with Pixel = back 2, the green flag sends both to 3 → SAME number → YES, they MEET.

Interaction note: the student's new script runs at the same time as Bit's; choosing back 2 makes the two concurrent results match.

Watch for & differentiate

Common error: moving Pixel forward (away from 3) — remind students Pixel starts ABOVE the target ($5 > 3$), so it must go back.

Accept any correct script: a child who writes 'back 1, back 1' also lands on 3 — that is fully correct.

Simplify: count the gap on the number path (5 to 3 is 2 squares back).

Extend: ask for a DIFFERENT Pixel script that still meets Bit (e.g. forward 0 is not allowed, but back 1 then back 1 works).

Success indicator: the child writes a Pixel move landing on 3 and confirms both meet on 3 under the green flag.